

The time-related column (such as, Time_start, Time_end....) is matlab time format.

1. The meaning of each file:

- a. The *_LR_meta.csv has **n+1** rows (**n** is the number of spectral sequences of the day, 1 is the top row, which means the index of columns (you need to **skip the first row when loading the data**)).
- b. The *_LR_CAL.csv should be a $(3648+1) * (3n)$ matrix, 3648 is the length of **wavelength**, 1 is the top row, 3n means (radiance_1, irradiance_before_1, irradiance_after_1, radiance_2, irradiance_before_2, irradiance_after_2, ..., radiance_n, irradiance_before_n, irradiance_after_n)
- c. The *_LR_REFL.csv should be a $(3648+1) * (2n)$ matrix, 2n means (reflectance_before_1, reflectance_after_1, reflectance_before_2, reflectance_after_2, ..., reflectance_before_n, reflectance_after_n)

2. Note: the "before" and "after" representing the closest irradiance (reflectance) before and after each time of measuring the radiance spectrum. There is an obvious difference between irradiance_before and irradiance_after when the incident radiation is unstable or weak (such as in the morning or evening or cloudy), however, they are close when the incident radiation is strong (see pic below).
3. I'm not sure which one is the best to be the "corresponding irradiance" to the radiance. Maybe use the average value of them? Gabriel could you please give your opinion and tell us how you choose the irradiance to retrieve SIF? And if there are any problem on my explanations, please correct me.



